STPM 2020 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. The electric field strength at a distance of 0.50 m from a point charge is 8.9×10^3 N C^{-1} and the field is in the direction towards the charge. What is the value of the point charge?

A -1.2×10^{-7} C B $+1.2 \times 10^{-7}$ C C -2.5×10^{-7} C D $+2.5 \times 10^{-7}$ C

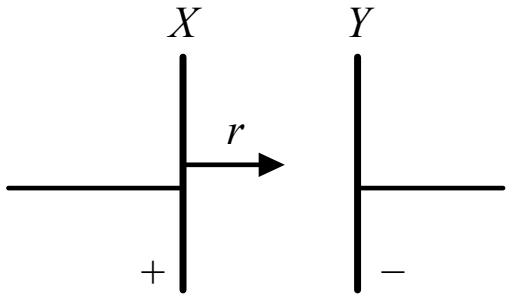
Answer: C

$$|Q| = \frac{Er^2}{k} = \frac{(8.9 \times 10^3)(0.50)^2}{9.0 \times 10^9} = 2.5 \times 10^{-7} \text{ C}$$

The field points towards the charge, so the charge is negative: -2.5×10^{-7} C.

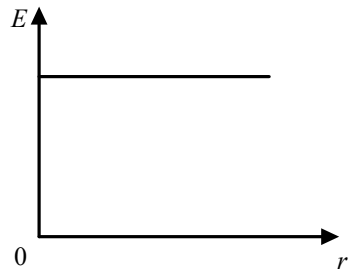
Explanation: Field lines point away from positive and towards negative charges.

2. Two parallel metal plates, *X* and *Y*, have opposite charges and separated at a fixed distance are shown in the diagram below.

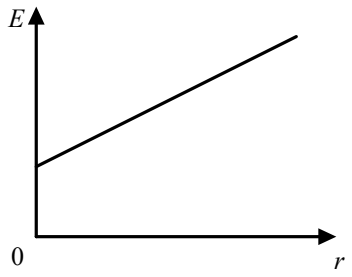


Which graph shows the variation of the magnitude of electric field strength, *E*, with distance, *r*, from *X*?

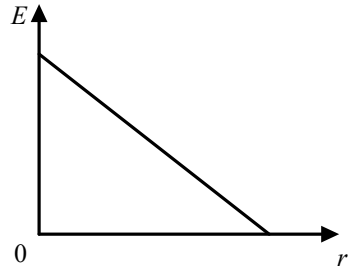
A



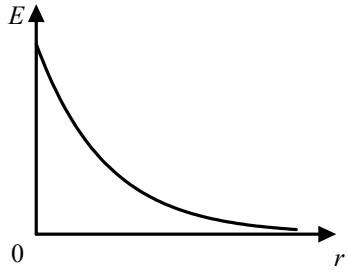
B



C



D



Answer: A

Between parallel plates the field is uniform: E is constant with distance (A).

Explanation: $E = V/d$ everywhere between the plates (away from the edges).

3. The capacitance of a parallel plate capacitor is

A independent of the dielectric in the capacitor
B inversely proportional to the area of the plates of the capacitor
C directly proportional to the distance between the plates of the capacitor
D independent of the potential difference across the plates of the capacitor

Answer: D

$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$: it depends only on the geometry and dielectric, not on the p.d.

Explanation: Doubling V doubles Q, so Q/V is unchanged.

4. A parallel plate capacitor is constantly charged by a battery. After several minutes of charging, a slab of dielectric material with dielectric constant, ε_r, is inserted in the space between the plates of the capacitor. The battery remains connected. What is the ratio of the energy stored before the slab is inserted to the energy stored after the slab is inserted?

A $1 : \varepsilon_r$ B $\varepsilon_r : 1$ C $1 : \varepsilon_r^2$ D $\varepsilon_r^2 : 1$

Answer: A

V fixed: $U = \frac{1}{2}CV^2 \propto C$; C becomes $\varepsilon_r C \Longrightarrow U_{\text{before}} : U_{\text{after}} = 1 : \varepsilon_r$

Explanation: With the battery connected the p.d. stays constant.

5. When a steady current flows in a non uniform wire, the current density is

A constant along the wire
B directly proportional to the radius at one point
C directly proportional to the drift velocity at one point
D inversely proportional to the density of free electrons

Answer: C

$J = nev_d$: at any point the current density is proportional to the drift velocity

Explanation: The current is the same everywhere, but the current density is larger where the wire is thinner.

6. A wire of resistance 21 Ω is melted and reshaped into a new wire of three times the length of the original wire. If the volume of the new wire is equal to the original wire, what is the resistance of the new wire?

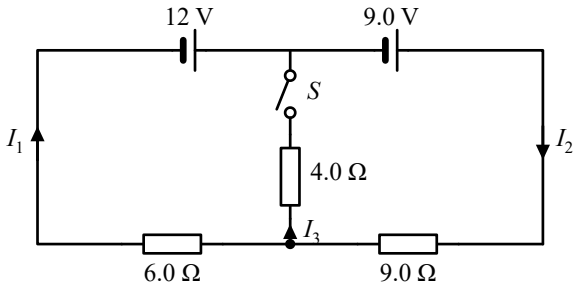
A 189 Ω B 63 Ω C 21 Ω D 7.0 Ω

Answer: A

Volume constant: length × 3 $\Longrightarrow$ area ÷ 3; $R = \frac{\rho l}{A} \Longrightarrow R \times 9 = 189\ \Omega$

Explanation: Both changes increase the resistance.

7. An electric circuit is set up as shown in the diagram below.



Which is **not** true about I_1, I_2 and I_3 when S is opened?

A $I_3 = 0$ B $I_1 = 0.20$ A C $I_1 = I_2$ D $I_2 = 1.4$ A

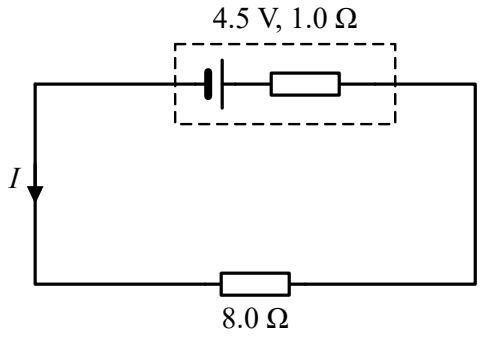
Answer: D

With S open no current flows in the 4.0 Ω branch: $I_3 = 0$ and $I_1 = I_2$ (single loop).

The cells oppose: $I = \frac{12 - 9.0}{6.0 + 9.0} = 0.20$ A, so $I_2 = 1.4$ A is not true

Explanation: In a single loop the same current flows everywhere.

8. A circuit of a battery connects to a resistor is shown in the diagram below.



What is the terminal potential difference of the battery?

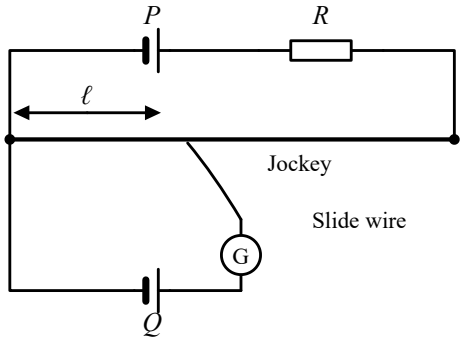
A 0.50 V B 3.5 V C 4.0 V D 4.5 V

Answer: C

$$I = \frac{4.5}{8.0 + 1.0} = 0.50 \text{ A; terminal p.d.} = E - Ir = 4.5 - 0.50 = 4.0 \text{ V}$$

Explanation: Equivalently, $V = IR = (0.50)(8.0)$.

9. A potentiometer, a device used to determine the e.m.f. of a cell, Q, has a balanced length, ℓ, is shown in the diagram below.



Which procedure would give a new balanced length that is longer than ℓ?

A A resistor is connected across Q.
B R is replaced with a resistor which has higher resistance.
C The terminal connection of P is reversed.
D Q is replaced with a cell which has smaller e.m.f.

Answer: B

A larger R reduces the current in the slide wire, so the p.d. per unit length falls; a longer length is needed to balance the same e.m.f.

Explanation: A resistor across Q does not change its e.m.f. at balance; reversing P gives no balance; a smaller e.m.f. gives a shorter length.

10. In Thomson's experiment to determine the ratio $\frac{e}{m}$ for electrons, the electron is launched with velocity v into a region consisting of crossed magnetic and electric fields. If the magnetic flux density is B and the electric field strength is E, which is true about the magnitude and direction of velocity of the electron?

	Magnitude	*Direction*
A	$\frac{B}{E}$	Parallel to the magnetic field
B	$\frac{E}{B}$	Parallel to the electric field
C	$\frac{B}{E}$	Perpendicular to the magnetic and electric fields
D	$\frac{E}{B}$	Perpendicular to the magnetic and electric fields

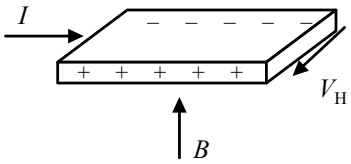
Answer: D

Undeflected: $eE = evB \Longrightarrow v = \frac{E}{B}$, perpendicular to both fields

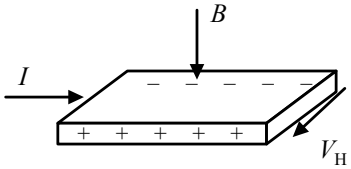
Explanation: Only perpendicular motion experiences both forces in opposite directions.

11. A current, I, flows in a conductor placed in a magnetic field of magnetic flux density, B. The Hall voltage, V_H, is produced across the sides of the conductor. Which diagram shows the correct direction of B?

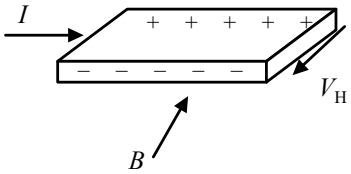
A



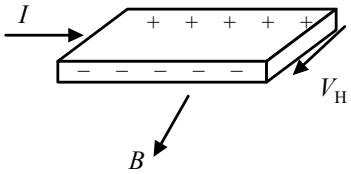
B



C



D



Answer: B

The Hall voltage appears across the faces perpendicular to both I and **B**. Applying Fleming's left-hand rule to the charge carriers, only diagram B gives the charge build-up shown on the faces.

Explanation: B must be perpendicular to the current and to the direction of V_H.

12. A solenoid is wound with 5000 turns on a hollow cardboard of radius 7.5 mm. If the length of the solenoid is 0.60 m and the current flows in the solenoid is 3.0 A, what is the magnetic flux through the solenoid?

A 1.0×10^{-2} Wb B 1.7×10^{-2} Wb C 2.8×10^{-2} Wb D 4.0×10^{-2} Wb

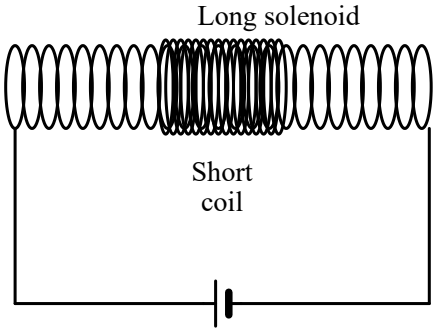
Answer: C

$$B = \frac{\mu_0 NI}{l} = \frac{(4\pi \times 10^{-7})(5000)(3.0)}{0.60} = 3.14 \times 10^{-2} \text{ T}$$

$$N\Phi = NBA = (5000)(3.14 \times 10^{-2})\pi(7.5 \times 10^{-3})^2 = 2.8 \times 10^{-2} \text{ Wb}$$

Explanation: The answer is the total flux linking all 5000 turns (flux linkage); the flux through one turn is 5.6×10^{-6} Wb.

13. A short coil of 600 turns is wound over the middle part of a long solenoid of 2000 turns as shown in the diagram below.



If the cross-sectional area of the long solenoid is 6.0×10^{-4} m^2 and the length of the solenoid is 0.75 m, what is the mutual inductance between the solenoid and the coil?

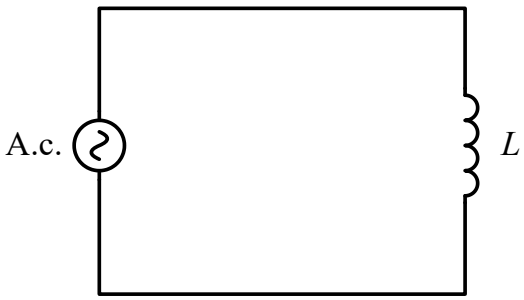
A 4.0×10^{-3} H B 1.2×10^{-3} H C 9.1×10^{-4} H D 3.6×10^{-4} H

Answer: B

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \frac{(4\pi \times 10^{-7})(2000)(600)(6.0 \times 10^{-4})}{0.75} = 1.2 \times 10^{-3} \text{ H}$$

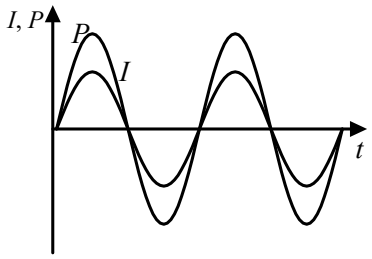
Explanation: The flux in the solenoid links every turn of the short coil.

14. A pure inductor, L, is connected to an alternating current source as shown in the diagram below.

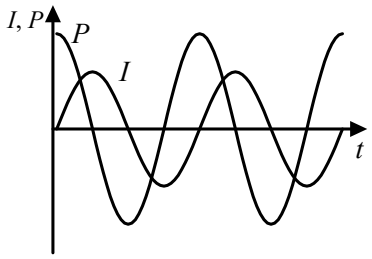


Which graph represents the variation of current, I, and power, P, with time, t, in the circuit?

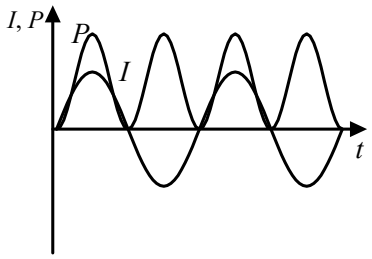
A



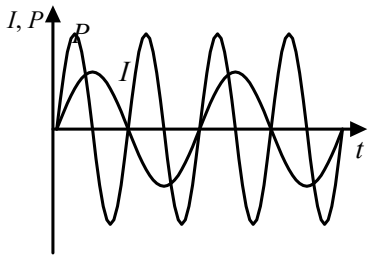
B



C



D

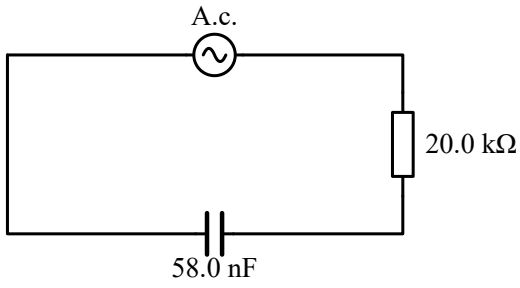


Answer: D

In a pure inductor the current lags the voltage by 90°, so $P = VI \propto \sin \omega t \cos \omega t$: it varies at twice the frequency and is positive and negative equally (D).

Explanation: The average power in a pure inductor is zero.

15. A resistor and a capacitor are connected in series to an alternating current source as shown in the diagram below.



If the frequency of the alternating current source is 60.0 Hz and the r.m.s. voltage across the resistor is 45.0 V, what is the r.m.s. voltage across the capacitor?

A 73.0 V B 103 V C 146 V D 647 V

Answer: B

$$I = \frac{45.0}{20.0 \times 10^3} = 2.25 \times 10^{-3}\ \text{A};\ X_C = \frac{1}{2\pi(60.0)(58.0 \times 10^{-9})} = 4.57 \times 10^4\ \Omega$$

$$V_C = IX_C = 103\ \text{V}$$

Explanation: The same current flows through both components in series.

Section B [15 marks]

16. A heating coil of 1.00 kW is designed to operate from a power supply of 240 V at room temperature. The coil is made of nichrome wire of diameter 0.50 mm.
[The resistivity of nichrome is 1.50×10^{-6} Ω m.]

(a) Calculate the length of the nichrome wire. [4]

Answer

$R = \frac{V^2}{P} = \frac{240^2}{1000} = 57.6\ \Omega$ 1

$A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}\ \text{m}^2$ 1

$l = \frac{RA}{\rho} = \frac{(57.6)(1.96 \times 10^{-7})}{1.50 \times 10^{-6}}$ 1

$l = 7.54$ m 1

4

Explanation: Use the radius, half the diameter.

(b) Explain the effect of the temperature increase on the power dissipated by the coil. [3]

Answer

As the temperature rises, the resistivity (and resistance) of the nichrome increases 1

The supply voltage is fixed, so $P = V^2/R$ 1

The power dissipated decreases 1

3

Total for question **7**

Explanation: The lattice vibrates more, increasing collisions with the electrons.

17. A galvanometer of resistance 25.0 Ω is converted into an ammeter using a shunt of resistance 1.00 Ω.

(a) The ammeter is used to measure the current in a circuit which consists of a resistor of resistance 15.0 Ω connected in series with a battery of 25.0 V. If the internal resistance of the battery is negligible, determine the current measured by the ammeter. [3]

Answer

Ammeter resistance $= \frac{(25.0)(1.00)}{25.0 + 1.00} = 0.962\ \Omega$ 1

$I = \frac{25.0}{15.0 + 0.962}$ 1

$I = 1.57$ A 1

3

Explanation: The ammeter adds a small resistance to the circuit.

(b) Determine the percentage difference between the measured current and the actual current. [3]

Answer

Actual current = $\frac{25.0}{15.0}$ = 1.667 A — 1

Percentage difference = $\frac{1.667 - 1.566}{1.667} \times 100\%$ — 1

= 6.0% — 1

3

Explanation: An ideal ammeter has zero resistance.

(c) Describe the function of shunt in the ammeter. [2]

Answer

The shunt is a low resistance in parallel with the galvanometer — 1

It carries most of the current so that only a small current passes through the galvanometer, allowing large currents to be measured — 1

2

Total for question **8**

Explanation: It also gives the ammeter a low overall resistance.

Section C [30 marks]

18. (a) Define *electric field strength at one point* in an electric field. [1]

Answer

The electric force per unit positive charge placed at that point ($E = F/q$) — 1

1

Explanation: Unit N C^{-1}; it is a vector.

(b) Explain the motion of an electron which moves into a uniform electric field

(i) in perpendicular direction to the electric field, [4]

Answer

The electron experiences a constant force eE opposite to the field	1
This gives a constant acceleration perpendicular to its initial velocity	1
Its velocity component perpendicular to the field is unchanged	1
So it follows a parabolic path, bending towards the positive plate	1
	4

Explanation: Like projectile motion under gravity.

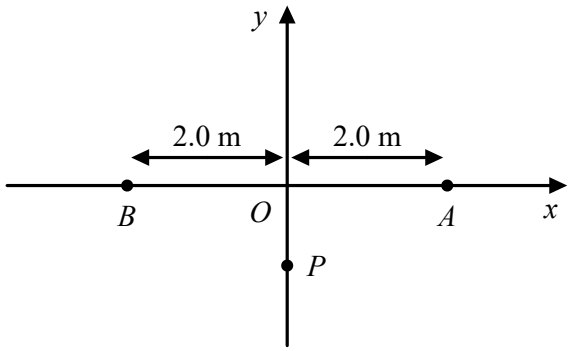
(ii) in the same direction as the electric field. [3]

Answer

The force on the electron is opposite to its velocity	1
so it decelerates uniformly in a straight line	1
comes momentarily to rest and then accelerates back in the opposite direction	1
	3

Explanation: The electron gains electric potential energy as it slows.

(c) Two equal point charges at A and B with the magnitude of 5.0 μC each are located opposites to each other 2.0 m from the origin, O, on the x-axis as shown in the diagram below.



Determine the magnitude and direction of the electric field strength at P, which is located 1.0 m below the origin, due to the point charges. [7]

Answer

Distance from each charge to $P = \sqrt{(2.0^2 + 1.0^2)} = 2.24$ m	1
E from each $= \dfrac{(9.0 \times 10^9)(5.0 \times 10^{-6})}{5.0}$	1
$= 9.0 \times 10^3$ N C^{-1}	1
Horizontal components are equal and opposite, so they cancel	1
Vertical components add: $E = 2(9.0 \times 10^3)\dfrac{1.0}{2.24}$	1
$E = 8.05 \times 10^3$ N C^{-1}	1
directed along $-y$ (downward, away from O) for positive charges	1
	7
Total for question	**15**

Explanation: Use symmetry: only the components along the y-axis survive.

19. (a)

(i) What is meant by *magnetic flux density*? [2]

Answer

The force per unit current per unit length on a conductor placed perpendicular to the magnetic field	1
$B = \dfrac{F}{IL}$; unit tesla (T)	1
	2

Explanation: 1 T = 1 N A^{-1} m^{-1}.

(ii) Explain the existence of a magnetic force acting on a conductor. [3]

Answer

A current is a flow of charges moving with a drift velocity	1
Each moving charge in the magnetic field experiences a force $qv_dB \sin\theta$	1
The sum of these forces on all the charges is the force on the conductor	1
	3

Explanation: The forces are transmitted to the lattice of the conductor.

(iii) A conductor of length L and cross-sectional area A carrying a current I is placed in a uniform magnetic field with magnetic flux density B at angle θ with the magnetic field. Derive an expression for the magnetic force on the conductor. [4]

Answer

Number of charge carriers in the conductor = nAL	1
Force on each = $qv_dB \sin\theta$	1
Total $F = nALqv_dB \sin\theta$	1
$I = nAqv_d \Longrightarrow F = BIL \sin\theta$	1
	4

Explanation: The microscopic and macroscopic descriptions agree.

(b) An electron is accelerated through a potential difference of 2500 V and then enters a region which has a uniform magnetic field of 0.25 T. Determine the maximum radius of the electron path in the magnetic field. [6]

Answer

$\frac{1}{2}mv^2 = eV$	1
$v = \sqrt{\left(\frac{2(1.60 \times 10^{-19})(2500)}{9.11 \times 10^{-31}}\right)}$	1
$v = 2.96 \times 10^7$ m s^{-1}	1
Maximum radius when $\mathbf{v} \perp \mathbf{B}$: $evB = \frac{mv^2}{r}$	1
$r = \frac{mv}{eB} = \frac{(9.11 \times 10^{-31})(2.96 \times 10^7)}{(1.60 \times 10^{-19})(0.25)}$	1
$r = 6.75 \times 10^{-4}$ m	1
	6
Total for question	**15**

Explanation: If the electron entered at an angle, the radius would be smaller (helical path).

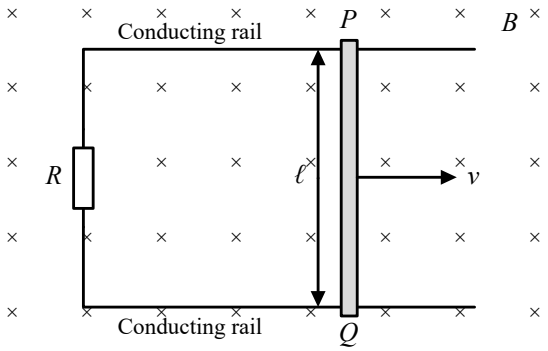
20. (a) State the Faraday's law and Lenz's law for electromagnetic induction. [2]

Answer

Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage	1
Lenz: the induced current flows in a direction that opposes the change producing it	1
	2

Explanation: Together: $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$.

(b) A metal rod PQ of length ℓ is moving at constant velocity, v, across a uniform magnetic field, B, on frictionless conducting rails which are connected by a resistor, R, as shown in the diagram below.



(i) Explain why a force is required to move the rod at constant velocity. [4]

Answer

As the rod moves, the flux through the circuit changes, so an e.m.f. is induced	1
A current flows in the rod and resistor	1
The current-carrying rod in the magnetic field experiences a force $BI\ell$ opposing the motion (Lenz)	1
An equal applied force is needed to keep the velocity constant (no resultant force)	1
	4

Explanation: Without the applied force the rod would slow down.

(ii) Derive an expression for the force by using the laws of electromagnetic induction. [7]

Answer

In time Δt the rod moves $v\Delta t$; area swept $\Delta A = \ell v \Delta t$	1
$\Delta\Phi = B\Delta A = B\ell v\Delta t$	1
$\varepsilon = \frac{\Delta\Phi}{\Delta t} = B\ell v$	1
$I = \frac{\varepsilon}{R} = \frac{B\ell v}{R}$	1
Magnetic force on the rod: $F = BI\ell$	1
$F = \frac{B^2\ell^2 v}{R}$	1
The applied force must equal this (in the direction of motion)	1
	7

Explanation: The work done by the applied force becomes heat in the resistor.

(iii) Determine the mechanical power to move the rod at constant speed of 5.0 m s^{-1} if ℓ is 1.2 m, R is 10.0 Ω and B is 2.0 T. [2]

Answer

$P = Fv = \frac{B^2\ell^2v^2}{R} = \frac{(2.0)^2(1.2)^2(5.0)^2}{10.0}$ 1

$P = 14.4$ W 1

2

Total for question **15**

Explanation: This equals the electrical power ε^2/R in the resistor.